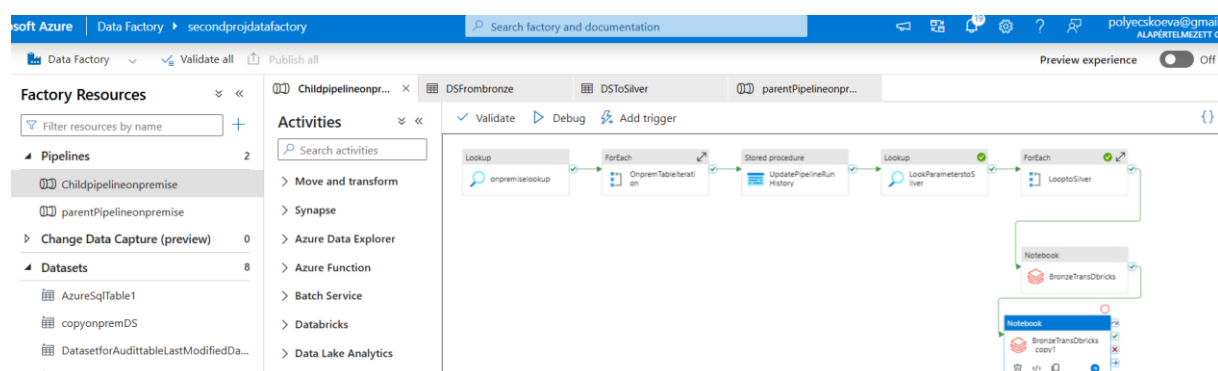


Table of Contents

A. Child Pipeline (main) – Childpipelineonpremise	1
1. Main Data Load from Prem SQL SERVER	1
2. Data Load from Bronze to Silver container	3
B. Parent pipeline - parentPipelineonpremise	3
1. Incremental load.....	4
2. Containers.....	5
C. Power Bi	7

A. Child Pipeline (main) – Childpipelineonpremise



Data Factory > secondprojdatafactory

Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Belgrade, Bratislava, ... : Last 24 hours Pipeline name: All Status: All Runs: Latest runs

Triggered by: All Add filter

Showing 1 - 9 items

☐	Pipeline name ↑↓	Run start ↑↓	Run end ↑↓	Duration	Triggered by	Status ↑↓	Run	Parameters
☐	Childpipelineonpremise	1/12/2025, 8:51:39 PM	1/12/2025, 8:57:50 PM	6m 11s	4b25d1d4-2e4b-40a...	✓ Succeeded	Original	
☐	parentPipelineonpremise	1/12/2025, 8:51:26 PM	1/12/2025, 8:57:53 PM	6m 28s	Manual trigger	✓ Succeeded	Original	
☐	Childpipelineonpremise	1/12/2025, 8:28:32 PM	1/12/2025, 8:37:41 PM	9m 9s	02396777-30e5-407...	✓ Succeeded	Original	
☐	parentPipelineonpremise	1/12/2025, 8:28:17 PM	1/12/2025, 8:37:44 PM	9m 27s	Manual trigger	✓ Succeeded	Original	

Last refreshed 0 minutes

1. Main Data Load from Prem SQL SERVER

Lookup Activity:

- The **Lookup Activity** is used to query the SQL database to retrieve the table names that need to be processed.

- The query result contains all the table names, which are then passed to the **For Each Activity** for further processing.

For Each Activity:

- The **For Each Activity** iterates through the values returned by the **Lookup Activity** (the table names).
- For each table, it triggers an embedded **Copy Activity**, which copies data from the source SQL database to the destination (Azure Data Lake Gen2).
- During each iteration, the **For Each** passes the current table name dynamically to the next step.

Copy Activity (Dynamic Settings):

- In the **Copy Activity**, the **Source Dataset** is dynamically configured to use the **Table Name** field, so the current table name from the iteration is used.
- In the **Sink Dataset**, I dynamically generated the target path where the Parquet files are stored: I'm writing each table's data into its own folder inside the **bronze** container.

Linked Services and Datasets:

- The **Linked Services** connect the source (SQL Server) and target systems (Azure Data Lake Gen2). The source Linked Service handles the connection to SQL Server, while the target Linked Service manages writing to the Data Lake.
- The **Self-hosted Integration Runtime** facilitates the data transfer from the local environment to Azure.
- The **SQL Server Dataset** is used for querying the tables, while the **Data Lake Gen2 Dataset** is used to write the Parquet files to the target location. The table names and file paths are configured dynamically.

Integration runtimes

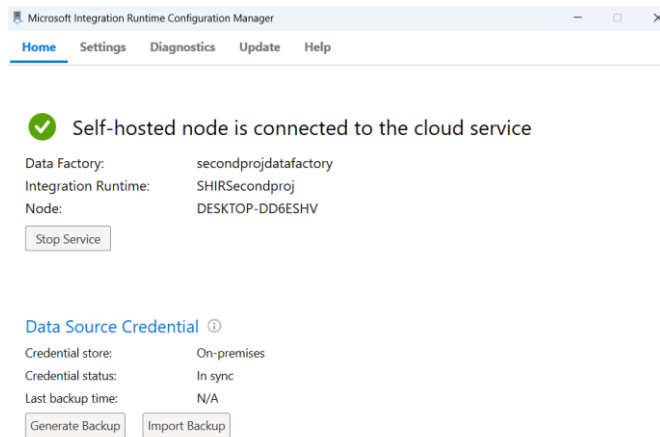
The integration runtime (IR) is the compute infrastructure to provide the following data integration capabilities across different network environment. [Learn more](#)

[+ New](#) [Refresh](#)

Showing 1 - 2 of 2 items

Name ↑↓	Type ↑↓	Sub-type ↑↓	Status ↑↓	Related ↑↓	Region ↑↓	Version ↑↓
 AutoResolveIntegrationR...	Azure	Public	 Running	0	Auto Resolve	---
 SHIRSecondproj	Self-Hosted	---	 Running	1	---	5.48.9106.2

On my computer:



2. Data Load from Bronze to Silver container

Copy Activity Datasets:

- Dynamically changing the directory and folder names in both the source and sink. The parameters are defined in the JSON file located in the parameter container.

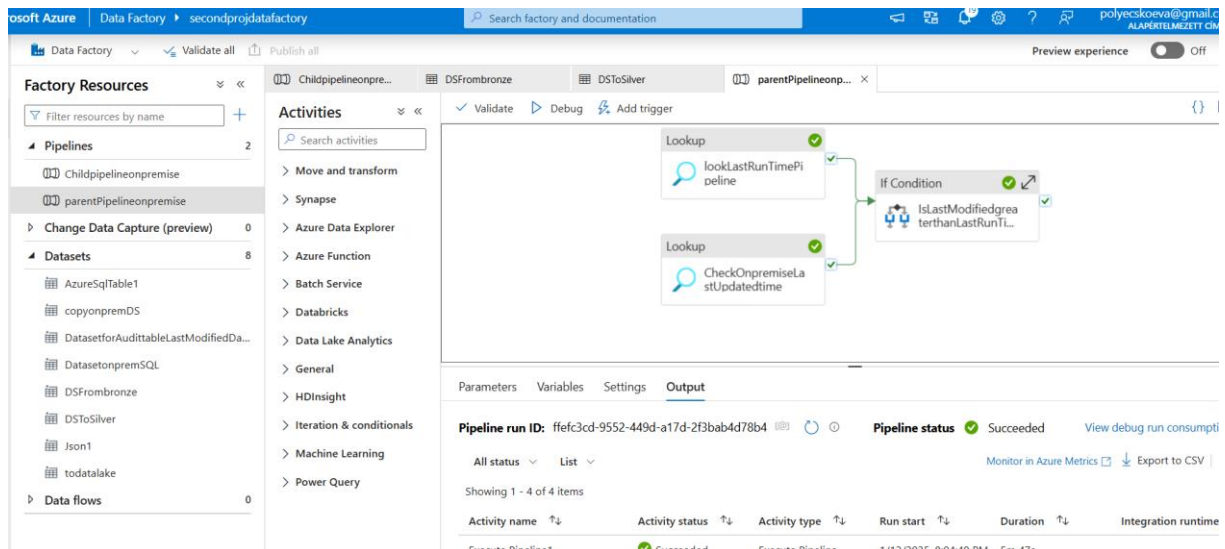
Lookup Activity:

- The **Lookup Activity** is used to look at the json file on the bronze container with 7 items. Each item is a dictionary, contains one table (e.g. Customers) parameter values.

For Each Activity:

- The **For Each Activity** iterates through the values returned by the **Lookup Activity**
- For each table, it triggers an embedded **Copy Activity**, which copies data to the destination silver container (Azure Data Lake Gen2).

B. Parent pipeline - parentPipelineonpremise

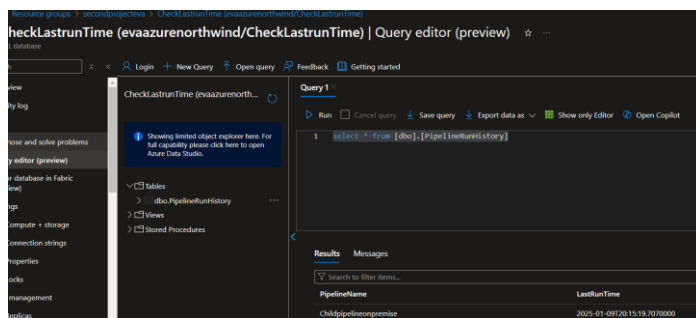


1. Incremental load

For this, I have created an Azure SQL database with one table representing the lastruntime.

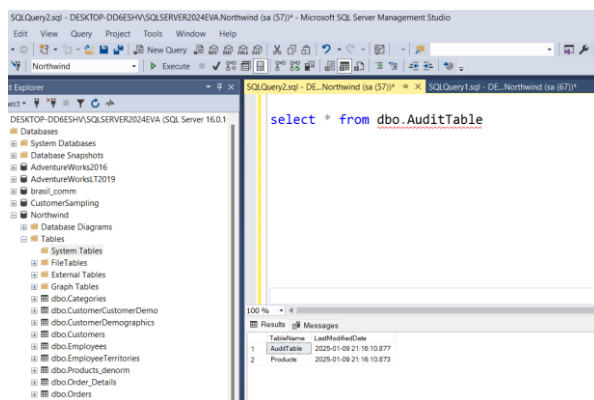
Lookup Activity for Azure SQL Database:

- A Lookup activity queries the PipelineRunHistory table in the Azure SQL Database to retrieve the most recent runtime date (LastRunTime) for the pipeline.



Lookup Activity for On-Premises SQL Server:

- Another Lookup activity retrieves the LastModifiedDate from the audit table in the on-premises SQL Server, representing the most recent modification in the source system.



Comparison Using If Condition Activity:

- An If Condition activity compares the two dates: the LastModifiedDate from the on-premises audit table and the LastRunTime from the Azure SQL Database.
- The comparison uses an expression like @greater() to check if the LastModifiedDate is more recent.
- Time zone adjustment is applied (subtracting an hour from the on-premises LastModifiedDate) to align timestamps before comparison.

Execution of Child Pipeline:

- If the LastModifiedDate is more recent, the If Condition activity triggers a child pipeline to process and load the new data.

Stored Procedure for Updating PipelineRunHistory:

- A stored procedure (UpdatePipelineRunHistory) is executed to update the PipelineRunHistory table in the Azure SQL Database with the current pipeline run details, including its completion status and timestamp.

No New Data Handling:

- If the LastModifiedDate is not more recent, the If Condition activity places a Set Variable activity to log that no new data is available for processing.
- The pipeline stops without running further steps.

Time Zone Adjustment:

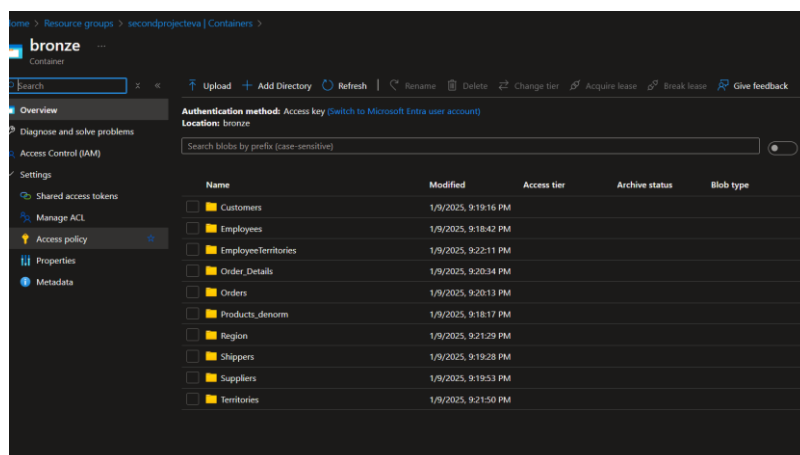
- A time zone adjustment has been applied (e.g., subtracting an hour from the on-premises LastModifiedDate) to align timestamps before comparison.

Efficient Incremental Processing:

- This setup ensures that the pipeline processes only new or updated data, optimizing resource usage and avoiding redundant data loads.

2. Containers

Bronze Container



Silver

secondprojecteva | Containers >

UploadAdd DirectoryRefreshRenameDeleteChange tierAcquire leaseBreak lease

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: silver

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status
☐ _azuretmpfolder\$	1/10/2025, 6:32:08 PM		
☐ Categories	1/12/2025, 8:56:22 PM		
☐ Customers	1/12/2025, 8:18:55 PM		
☐ Employees	1/12/2025, 8:56:25 PM		
☐ EmployeeTerritories	1/12/2025, 8:18:56 PM		
☐ Order_Details	1/12/2025, 8:18:56 PM		
☐ Orders	1/12/2025, 8:56:30 PM		
☐ Products_denorm	1/12/2025, 8:56:28 PM		
☐ Region	1/12/2025, 8:18:59 PM		
☐ Shippers	1/12/2025, 8:18:56 PM		
☐ Suppliers	1/12/2025, 8:18:58 PM		
☐ Territories	1/12/2025, 8:18:57 PM		

Gold

secondprojecteva | Containers >

UploadAdd DirectoryRefreshRenameDeleteChange tierAcquire leaseBreak lease

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: gold

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status
☐ _azuretmpfolder\$	1/12/2025, 8:50:44 PM		
☐ Categories	1/12/2025, 8:57:07 PM		
☐ Customers	1/12/2025, 8:57:22 PM		
☐ Employees	1/12/2025, 8:57:24 PM		
☐ EmployeeTerritories	1/12/2025, 8:57:19 PM		
☐ Order_Details	1/12/2025, 8:57:16 PM		
☐ Orders	1/12/2025, 8:57:04 PM		
☐ Products_denorm	1/12/2025, 8:57:26 PM		
☐ Region	1/12/2025, 8:57:14 PM		
☐ Shippers	1/12/2025, 8:57:11 PM		
☐ Suppliers	1/12/2025, 8:57:02 PM		
☐ Territories	1/12/2025, 8:57:09 PM		

Parameters

secondprojecteva | Containers >

UploadAdd DirectoryRefreshRenameDeleteChange tierAcquire leaseBreak lease

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: parameters

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier
☐ parameterstoSilver.json	1/12/2025, 7:05:41 PM	Hot (Inferred)

topowerbi

secondprojecteva | Containers >

UploadAdd DirectoryRefreshRenameDeleteChange tierAcquire leaseBreak lease

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: topowerbi

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Archive status
☐ _azuretmpfolder\$	1/12/2025, 9:23:49 PM		
☐ _committed_8236880057411830936	1/12/2025, 9:36:49 PM	Hot (Inferred)	
☐ _started_8236880057411830936	1/12/2025, 9:36:48 PM	Hot (Inferred)	
☐ _SUCCESS	1/12/2025, 9:36:49 PM	Hot (Inferred)	
☐ part-00000-fid-8236880057411830936-214854c5-...	1/12/2025, 9:36:48 PM	Hot (Inferred)	

C. Power Bi

